Σχεδιασμός Βάσεων Δεδομένων
Διδάσκων: Ιωάννης Κωτίδης

Εαρινό εξάμηνο 2020-2021

Δεύτερη Εργασία

Ανάθεση: **24-05-2021**
Παράδοση: **06-06-2020 Ώρα (23:55)**

Οδηγίες

- *Η εργασία είναι ατομική και υποχρεωτική.*
- *Η υποβολή της εργασίας πρέπει να γίνει στο eclass.*
- *Το παραδοτέο σας θα πρέπει να είναι ένα αρχείο PDF με όνομα AM.pdf (όπου AM είναι ο αριθμός μητρώου σας. π.χ. "3170001.pdf").*
- *Πιθανή αντιγραφή θα τιμωρείται με μηδενισμό όλων των εμπλεκομένων.*

Δημιουργία Αποθήκης Δεδομένων

ΑΠΑΝΤΗΣΕΙΣ

Ζήτημα 1 - Λύση

```
/* create table campdata and load data */

Create  table campdata
(custID int,
 fname varchar(30),
 lname varchar(30),
 cID   int,
 country varchar(30),
 bookID int,
 bookDate date,
 campCode char(3),
 CampName varchar(50),
 empno int,
 catCode char(1),
 category varchar(20),
 unitCost numeric(4,2),
 startDate date,
 overnights int,
 persons    int
 );

BULK INSERT campdata
FROM 'C:\data\campdata.txt'
WITH (FIRSTROW =2, FIELDTERMINATOR='|', ROWTERMINATOR = '\n');

/* crerate DW schema */

Create table customers
  (custId int primary key,
   fname varchar(30),
   lname varchar(30)
   );
insert into customers select distinct custID,fname,lname from campdata

create table countries
(cid int primary key,
 country varchar(30)
 );
insert into countries select distinct cid,country from campdata

create table campings
( campCode char(3) primary key,
  campName varchar(30)
);
insert into campings select distinct campCode, campName from campdata

create table categories
(catCode char(1) primary key,
 category varchar(10)
 );

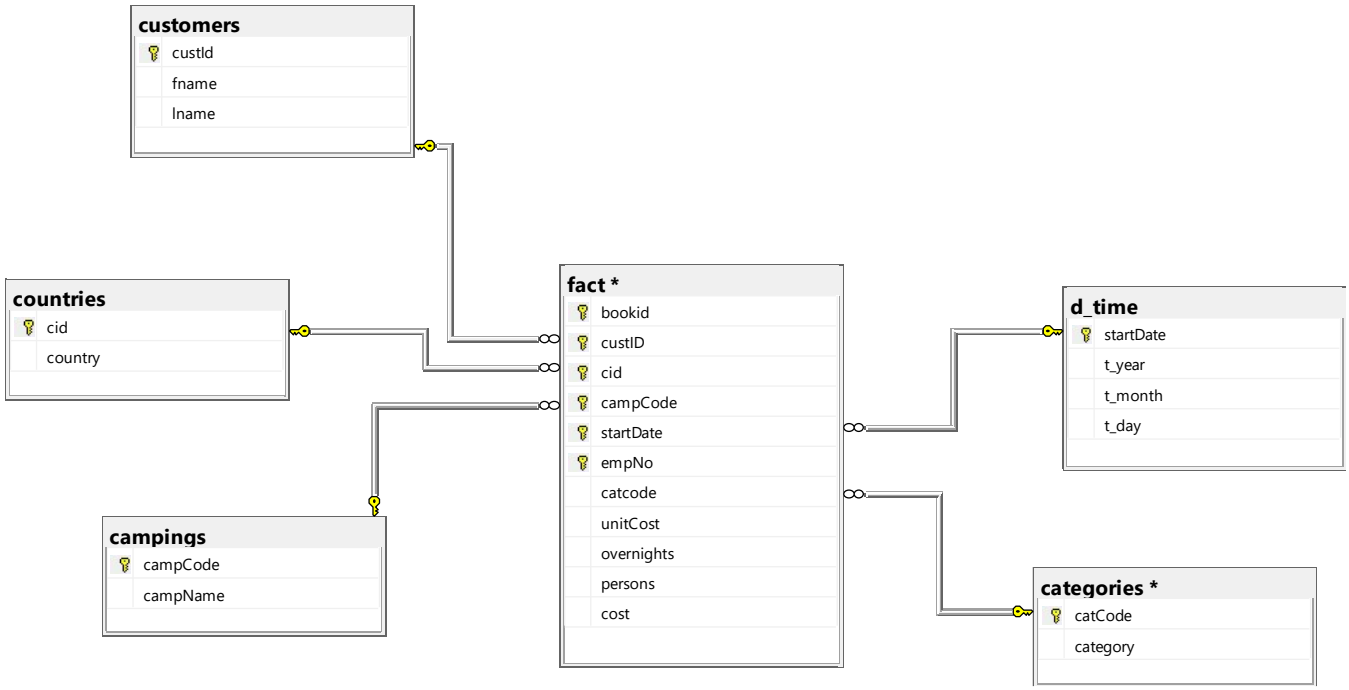
 insert into categories select distinct catCode,category from campdata
```

```
create table d_time
 (
   startDate date primary key,
   t_year int,
   t_month int,
   t_day int
)

insert into d_time
    select distinct startDate, datepart(year, startDate),
           datepart(month, startDate), datepart(day, startDate)
                from campdata

create table fact
(bookid int,
 custID int foreign key references customers(custID),
 cid int foreign key references countries (cid),
 campCode char(3) foreign key references campings (campCode),
 startDate date foreign key references d_time(startDate),
 empNo int,
 catcode char(1) foreign key references categories(catCode),
 unitCost numeric(4,2),
 overnights int,
 persons int,
 cost numeric (6,2)
 primary key (bookid, custID, cid, campCode, startDate, empNo)
);

insert into fact
select bookid, custID, cid, campCode, startDate, empNo, catcode,
unitcost,overnights, persons, (overnights*persons*unitcost)
 from campdata
```



Ζήτημα 2 - Λύση

```
--Q1

select top 100 country,lname, fname, sum(cost)
    from customers, countries, fact
 where customers.custid=fact.custid and
       countries.cid=fact.cid
  group by country,lname, fname
  order by sum(cost) desc

--Q2
select campname, category, sum(cost)
   from campings,categories, d_time, fact
  where campings.campCode=fact.campCode and
        d_time.startDate=fact.startDate and
        categories.catcode=fact.catCode and
              t_year = 2000
        group by campname,category
        order by campname,category

 --Q3
 select campname, t_month, sum(cost)
    from campings, d_time, fact
  where campings.campcode=fact.campcode and
        d_time.startDate=fact.startDate and
              t_year = 2018
              group by t_month,campname
--Q4
select t_year, campName, category, sum(persons)
   from d_time, campings, categories, fact
  where d_time.startDate=fact.startDate and
        campings.campCode = fact.campcode and
              categories.catCode = fact.catCode
  group by ROLLUP (t_year, campName, category)

--Q5
CREATE VIEW V2017 AS
   select campings.campcode, campname, sum(persons) as numofpersons
       from campings, d_time, fact
       where campings.campCode=fact.campCode and
             d_time.startDate=fact.startDate and
                t_year=2017
       group by campings.campCode, campname

CREATE VIEW V2018 AS
   select campings.campcode, campname, sum(persons) as numofpersons
       from campings, d_time, fact
       where campings.campCode=fact.campCode and
             d_time.startDate=fact.startDate and
                t_year=2018
       group by campings.campCode, campname


select v2018.campname
         from v2018, v2017
        where v2018.campcode=v2017.campcode and
              v2018.numofpesons > v2017.numofpersons
```

Ζήτημα 4 - Λύση

row#	Austria	Belgium	Denmark	Finland	France	Germany	Italy	Netherlands	Spain	Sweeden
69	0	0	0	0	0	0	0	0	1	0
70	0	0	0	0	0	0	0	0	1	0
71	0	0	1	0	0	0	0	0	0	0
72	1	0	0	0	0	0	0	0	0	0
73	1	0	0	0	0	0	0	0	0	0
74	0	0	0	0	0	1	0	0	0	0
75	0	0	0	0	0	1	0	0	0	0
76	0	0	0	0	0	1	0	0	0	0